

Task-Identity: Detecting Behavioral Drift Beyond Accuracy and Embedding Similarity

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Abstract

We introduce Task-Identity, a training-free metric for detecting behavioral drift in classification models, computed as the Pearson correlation between flattened confusion matrices from two evaluation conditions. Unlike accuracy monitoring, which tracks aggregate performance, Task-Identity measures whether a model is making the same pattern of mistakes. Unlike embedding-based drift detection, which measures structural similarity in representation space, Task-Identity measures similarity in decision behavior. These two distinctions matter in practice.

In our tested sample, a model maintained 93.6% to 93.7% accuracy while Task-Identity detected a 42.4% behavioral shift caused by class imbalance. In a separate experiment, embedding similarity remained at 0.583 during a complete model collapse to 0% accuracy on the original task, while Task-Identity correctly read 0.000, a gap of 58.3 percentage points. Per-class Task-Identity analysis revealed a targeted poisoning attack on two specific classes that an aggregate score of 0.873 had concealed. In a post-filing financial lending extension using 2.26 million real loans, an overall score of 0.921 masked a 99.4% degradation in minority class detection.

We validate Task-Identity across 11 tests spanning four domains: computer vision, natural language processing, medical AI, and audio classification. All tests use real published datasets with no synthetic data. We report explicit negative results alongside positive findings, including a progressive noise experiment where accuracy and Task-Identity moved together rather than diverging.

1 Introduction

Deployed machine learning classifiers are often monitored using accuracy or other aggregate performance metrics. When accuracy drops, something is wrong. When accuracy holds steady, the usual assumption is that the model is still behaving normally. This paper presents evidence that this assumption can fail in ways that matter for real classification systems.

We study behavioral drift in classification models using a scalar metric called Task-Identity, computed as the Pearson correlation between flattened confusion matrices from two evaluation conditions. The metric measures whether a model is making the same pattern of mistakes, not whether it is achieving the same overall score. This distinction turns out to be important. A model can maintain stable accuracy while its decision behavior shifts substantially, and a model can appear structurally similar to a baseline by embedding-based measures while its actual predictions have collapsed entirely.

We demonstrate both failure modes empirically. In one case, a model maintained 93.6% to 93.7% accuracy while Task-Identity detected a 42.4% behavioral shift caused by class imbalance.

Traditional accuracy monitoring would have flagged nothing. In another, embedding similarity remained at 0.583 during a complete model collapse to 0% accuracy on the original task, while Task-Identity correctly read 0.000. The gap between what embedding similarity reported and what was actually happening was 58.3 percentage points.

We validate Task-Identity across 11 tests spanning four domains: computer vision, natural language processing, medical AI, and audio classification. All tests use real published datasets with no synthetic data. We report results that did not work as expected alongside the ones that did.

This paper makes the following contributions:

- We show that accuracy monitoring can miss substantial behavioral drift, demonstrated empirically with a 42.4% behavioral shift invisible to accuracy in our tested sample.
- We show that embedding similarity and Task-Identity measure fundamentally different things and can diverge sharply during catastrophic failure, with a 58.3 percentage point gap in our tested sample.
- We show that aggregate Task-Identity scores can conceal catastrophic class-level failures in imbalanced settings, and that per-class analysis can recover the hidden behavioral collapse.
- We validate Task-Identity across four domains using the same confusion matrix method with no domain-specific modification.

All experiments use real published datasets with no synthetic data. We report explicit negative results alongside the positive findings.

2 The Task-Identity Framework

2.1 Motivation

Standard drift detection methods fall into two broad categories. Data drift methods monitor the input distribution, checking whether the data arriving at a model has shifted from what the model was trained on. Performance drift methods monitor output metrics like accuracy, F1, or AUC over time. Both approaches have a shared limitation: neither directly measures what the model is doing with its inputs. A model that receives the same distribution of inputs and produces the same aggregate accuracy score can still be making systematically different decisions, confusing different classes, and exhibiting entirely different failure patterns.

Task-Identity addresses this gap by measuring the model’s behavioral fingerprint directly, using the confusion matrix as its representation of what the model does.

2.2 The Metric

For a classification model evaluated on a labeled dataset, the confusion matrix captures the model’s observed pattern of correct predictions and errors across the class set. Two models, or the same model under two different evaluation conditions, can be compared by asking: do their confusion matrices look alike?

Task-Identity answers this question using the Pearson correlation between the two matrices after flattening them to one-dimensional vectors:

$$\text{Task-Identity} = \max(0, \text{Pearson}(\text{flatten}(CM_1), \text{flatten}(CM_2))) \quad (1)$$

where CM_1 is the confusion matrix from the baseline evaluation condition, CM_2 is the confusion matrix from the current evaluation condition, and the $\max(0, \cdot)$ floor prevents negative correlations from being reported as meaningful anti-similarity.

The result is a score in $[0.0, 1.0]$:

- A score near 1.0 means the model is exhibiting a highly similar pattern of mistakes in both conditions. The overall confusion structure is preserved even if absolute counts differ.
- A score near 0.0 means the model’s decision behavior has changed substantially. The pattern of errors is different, even if overall accuracy appears similar.

Task-Identity assumes that both evaluation conditions use the same label space and that confusion matrices are computed over comparable class sets.

2.3 Per-Class Extension

For settings with class imbalance, a single aggregate Task-Identity score can be dominated by the majority class and miss catastrophic drift in minority classes. The per-class extension computes Task-Identity separately for each class by isolating the class-specific confusion behavior and comparing it across conditions.

This allows detection of class-specific failures that aggregate scores conceal. We demonstrate this in a targeted poisoning experiment, where the overall Task-Identity score was 0.873 while the poisoned classes fell to approximately 0.17, and in a later financial lending extension where an overall score of 0.921 masked a complete collapse in minority class detection.

2.4 What Task-Identity Measures and What It Does Not

Task-Identity measures behavioral similarity: whether a model produces the same pattern of decisions across two evaluation conditions. It does not measure performance. A model with Task-Identity of 1.0 may be performing well or poorly; it is simply behaving consistently. A model with Task-Identity near 0.0 has changed its decision behavior substantially, which may or may not coincide with a change in accuracy.

This distinction is central to the findings in this paper. In the class imbalance experiment, accuracy was stable while Task-Identity detected a 42.4% behavioral shift. In the label space divergence experiment, the model completely forgot its original task while an embedding-based measure suggested moderate structural similarity.

Task-Identity also does not require access to model internals, making it applicable to classification systems without requiring access to model weights, activations, or internal representations.

3 Baseline Validation: Task-Identity on Near-Stable and Converged Models

3.1 Setup

Before applying Task-Identity to failure detection, we first check whether the metric behaves sensibly on models that are not undergoing major behavioral change. If Task-Identity produced noisy or unstable scores on models trained on the same task under nearby training conditions, it would be difficult to interpret low scores as evidence of drift. We therefore use the MNIST training-dynamics

experiment as a baseline check, comparing models trained on the same dataset with the same architecture for 5, 20, and 50 iterations. The dataset is MNIST handwritten digits (LeCun et al., 1998), with 7,000 training samples and 3,000 test samples.

3.2 Results

Comparison	Task-Identity	Interpretation
Undertrained (5 iter) vs Normal (20 iter)	0.999	Nearly identical
Undertrained (5 iter) vs Extended (50 iter)	0.999	Nearly identical
Normal (20 iter) vs Extended (50 iter)	1.000	Converged

Table 1: Task-Identity scores comparing models trained for different numbers of iterations on MNIST. Accuracy progression: 0.921 (5 iter) $\rightarrow$ 0.936 (20 iter) $\rightarrow$ 0.938 (50 iter).

Task-Identity of 1.000 between the 20-iteration and 50-iteration models indicates behavioral convergence: additional training changed accuracy by only 0.2 percentage points and produced no detectable change in decision behavior. The similarly high scores for the 5-iteration comparisons suggest that in this MNIST setup the model’s confusion pattern was already largely fixed early in training.

3.3 What This Shows

In this sample, Task-Identity produces scores near 1.0 across models trained on the same task at different training stages. The metric does not produce spurious low scores in this stable-task comparison. This provides a baseline against which the failure-detection results in later sections can be interpreted.

We note that these results are on a single dataset with a single architecture type. Whether Task-Identity is equally well-behaved on stable models across other settings is not established by this baseline alone.

4 Accuracy Fails to Detect Behavioral Drift

4.1 Setup

The most widely used signal for detecting model problems in production is accuracy. When accuracy drops, something has changed. When accuracy holds steady, the standard assumption is that the model is still behaving normally. We test whether this assumption holds under class imbalance.

We trained a neural network classifier on a balanced MNIST dataset (LeCun et al., 1998) with 7,000 training samples and 3,000 test samples. We then evaluated the model on two test sets: a balanced test set with equal class representation, and an imbalanced test set in which class 0 comprised 310 samples (51.1%) and classes 1 through 9 comprised 33 samples each (5.4% each). We computed accuracy and Task-Identity on both evaluations.

4.2 Results

Accuracy moved from 0.936 to 0.937, a change of 0.001 in absolute accuracy (0.1 percentage points). Task-Identity moved from 1.000 to 0.576, indicating a 42.4% behavioral shift. A system monitoring

Evaluation Condition	Accuracy	Task-Identity
Balanced test set	0.936	1.000 (reference)
Imbalanced test set	0.937	0.576

Table 2: Accuracy and Task-Identity under balanced and imbalanced evaluation conditions on MNIST. Accuracy changed by 0.001 in absolute accuracy (0.1 percentage points) while Task-Identity detected a 42.4% behavioral shift.

only accuracy would have flagged nothing. Task-Identity detected a substantial change in the model’s confusion pattern across the evaluation conditions.

4.3 What This Shows

In this sample, accuracy and Task-Identity conveyed completely different information about the same model under the same conditions. The behavioral shift detected by Task-Identity reflects a real change in the model’s confusion pattern: the model made different kinds of mistakes on the imbalanced distribution even though the overall accuracy score remained flat.

This result does not show that accuracy is always uninformative. It shows that accuracy can be uninformative in settings with class imbalance, where majority-class performance dominates the aggregate score.

5 Embedding Similarity and Task-Identity Measure Different Things

5.1 Setup

Embedding-based drift detection methods measure how similar a model’s internal representations are across two evaluation conditions. A common assumption is that structural similarity in embedding space implies behavioral similarity in prediction space. We test whether this assumption holds during catastrophic model failure.

We trained a neural network on MNIST digits 0 through 4, then retrained the same model on digits 5 through 9, creating a label-space divergence and catastrophic-forgetting condition for the original task. Embedding similarity was computed using the cosine similarity between the mean L2-normalized embedding vectors of the two model states, reported here as a comparison signal rather than as the proposed metric. Task-Identity was computed using the confusion matrix correlation method described in Section 2.

5.2 Results

The model successfully learned the new digit set after retraining, reaching 0.978 accuracy on the new task. At the same time, it lost all accuracy on its original task. Embedding similarity reported 0.583, suggesting moderate structural similarity between the two model states. Task-Identity reported 0.000, correctly detecting complete behavioral collapse on the original task. The gap between the two signals was 58.3 percentage points.

5.3 What This Shows

Embedding similarity and Task-Identity measure fundamentally different properties of a model. Embedding similarity captures structural similarity in representation space. Task-Identity captures

Metric	Value
Accuracy on original task before retraining	0.993
Accuracy on new task after retraining	0.978
Accuracy on original task after retraining	0.000
Embedding similarity	0.583
Task-Identity	0.000
Detection gap	58.3 percentage points

Table 3: Embedding similarity vs Task-Identity during catastrophic forgetting on MNIST. The model successfully learned the new task (0.978 accuracy) while completely forgetting the original task (0.000 accuracy). Embedding similarity remained at 0.583 while Task-Identity correctly read 0.000.

similarity in decision behavior. These two properties can diverge sharply during catastrophic failure, as shown here. In this sample, relying on embedding similarity alone would have missed a complete model failure by 58.3 percentage points.

This finding does not show that embedding similarity is always unreliable. It shows that embedding similarity and behavioral identity are not the same signal, and that in failure scenarios where decision behavior collapses while structural representations remain partially intact, the two metrics can deliver contradictory readings.

6 Per-Class Analysis Reveals Hidden Behavioral Collapse

6.1 Setup

A single aggregate Task-Identity score summarizes behavioral similarity across the full confusion matrix. In settings with class imbalance or targeted interventions, this aggregate can conceal catastrophic drift in specific classes while appearing stable overall. We demonstrate this in two experiments: a targeted data poisoning attack and a financial lending extension with extreme class imbalance.

6.1.1 Targeted Poisoning

We trained a neural network on MNIST (LeCun et al., 1998) with 7,000 training samples and 3,000 test samples. We then created a poisoned version of the training data by relabeling 60% of class 5 samples as class 3 and 60% of class 8 samples as class 3, and retrained the model on the poisoned data. We computed both overall Task-Identity and per-class Task-Identity for each of the 10 digit classes.

6.1.2 Financial Lending Extension

As a post-filing extension, we applied Task-Identity to a credit risk classification setting using the Lending Club 2007–2018 dataset (Lending Club, 2018), containing 2,260,701 real loan records with extreme class imbalance: 86.9% of loans were paid and 13.1% defaulted. We compared a baseline lending model against a drifted lending model and computed both overall and per-class Task-Identity scores.

6.2 Results

6.2.1 Targeted Poisoning

Class	Task-Identity	Status
0	1.000	Stable
1	1.000	Stable
2	1.000	Stable
3	1.000	Stable
4	1.000	Stable
5	0.171	Severe drift
6	1.000	Stable
7	1.000	Stable
8	0.177	Severe drift
9	1.000	Stable
Overall	0.873	Moderate (misleading)

Table 4: Per-class Task-Identity after a targeted poisoning attack on MNIST classes 5 and 8. Overall Task-Identity of 0.873 appears moderate. Per-class analysis reveals severe compromise (0.171 and 0.177) in exactly the two poisoned classes while all other classes remain at 1.000.

The overall Task-Identity of 0.873 suggested a moderate behavioral change. Per-class analysis revealed a completely different picture: the two poisoned classes showed severe behavioral collapse (0.171 and 0.177) while the eight unpoisoned classes remained at exactly 1.000. The poisoning attack was precisely localized to the targeted classes, and the overall score had concealed it.

The attack’s practical impact was substantial. On class 5, the clean model correctly classified 231 of 253 test samples (91%), while the poisoned model correctly classified only 31 (12%), with 215 of 253 samples misclassified as class 3. On class 8, the clean model correctly classified 254 of 294 samples (86%), while the poisoned model correctly classified only 37 (13%).

6.2.2 Financial Lending Extension

Metric	Value
Overall Task-Identity	0.921
Paid class Task-Identity	0.949
Default class Task-Identity	0.000
Default detection rate (baseline)	81.8%
Default detection rate (drifted)	0.5%
Degradation in default detection	99.4%

Table 5: Per-class Task-Identity on Lending Club 2007–2018 (2,260,701 loans). Overall Task-Identity of 0.921 masked a complete collapse in minority class (default) detection.

The overall Task-Identity of 0.921 suggested stable model behavior. Per-class analysis showed that the paid class was stable (0.949) while the default class had completely collapsed (0.000). The drifted model detected only 0.5% of defaults compared to 81.8% for the baseline, a 99.4% degradation in the class that matters most for credit risk.

6.3 What This Shows

In both experiments, aggregate Task-Identity scores concealed catastrophic class-level failures. In the poisoning case, an overall score of 0.873 masked a targeted attack on two specific classes that reduced their detection rates by roughly 80 percentage points. In the lending case, an overall score of 0.921 masked a near-total collapse in minority class detection.

These results suggest that per-class analysis can be necessary in imbalanced or targeted-intervention settings. An aggregate score that appears moderate or stable can be consistent with catastrophic failure in a specific class. This is particularly relevant in high-stakes domains where minority class performance carries disproportionate consequence, such as fraud detection, medical diagnosis, and credit risk.

7 Cross-Domain Validation

7.1 Setup

Task-Identity uses the same confusion matrix correlation method regardless of the domain or data type. We test this claim by applying the metric across four domains using no domain-specific modification: computer vision, natural language processing, medical AI, and audio classification. Each domain test uses a real published dataset.

The cross-domain tests follow a common structure: train a baseline model, apply an intervention that induces behavioral drift, and compute Task-Identity between the baseline and post-intervention model. The intervention in each case is catastrophic forgetting via imbalanced fine-tuning or single-class retraining.

7.2 Results

Domain	Dataset	Baseline Acc.	Post-Intervention Acc.
Computer Vision	MNIST (LeCun et al., 1998)	0.993	0.000
NLP	20 Newsgroups (Lang, 1995)	0.943	0.505
Medical AI	Wisconsin Breast Cancer (Wolberg et al., 1994)	0.971	0.374
Audio	Free Spoken Digit Dataset (Jackson et al., 2018)	0.938	0.000

Table 6: Task-Identity across four domains under catastrophic forgetting or severe imbalanced fine-tuning. The same confusion matrix method detects behavioral collapse in each domain with no domain-specific modification.

In this sample, Task-Identity registered the induced behavioral drift in each domain. In three of the four cases the score reached 0.000, indicating complete behavioral collapse. In the NLP case, the score reached 0.036, indicating near-total collapse consistent with a model that had collapsed to single-class prediction after imbalanced fine-tuning with a 10:1 class ratio.

7.3 What This Shows

The same confusion matrix correlation method produced meaningful behavioral drift signals across computer vision, text, medical, and audio classification settings without any domain-specific tuning. The underlying data types differed substantially: pixel arrays, TF-IDF text features, clinical

measurements, and MFCC audio features. Task-Identity operated on model predictions and true labels in all cases, requiring no knowledge of the underlying representation.

We note that all four cross-domain tests involved severe interventions producing large behavioral shifts. Whether Task-Identity distinguishes more subtle behavioral changes consistently across domains is not established by these results.

8 Additional Validation

8.1 Model Compression

Model compression reduces model size for deployment on edge devices, but can also introduce behavioral changes that require more than aggregate accuracy to characterize. We validated Task-Identity as a compression quality gate by comparing a full-precision MNIST model against a 4x compressed version simulating 8-bit quantization.

Model	Accuracy	Task-Identity
Full precision	0.936	1.000 (reference)
4x compressed	0.395	0.384

Table 7: Task-Identity between full-precision and 4x compressed MNIST model. Compression destroyed behavioral integrity across 6 of 10 classes while reducing accuracy by 54.1 percentage points.

Per-class analysis revealed that compression was not uniformly harmful. Classes 1 and 9 showed low impact, while classes 3, 4, 5, 6, 7, and 8 were effectively destroyed, with per-class accuracy dropping to near zero. Overall Task-Identity of 0.384 flagged the compressed model as behaviorally degraded and therefore unsuitable for deployment without further review.

8.2 Domain Shift

We tested whether Task-Identity detects when a model is applied to data from a different distribution than its training set. A model trained on MNIST digit images (LeCun et al., 1998) was evaluated on Fashion-MNIST clothing images (Xiao et al., 2017). The two datasets share identical format (28x28 grayscale images, 10 classes) but differ in semantic content.

Evaluation Condition	Accuracy	Task-Identity
In-distribution (MNIST)	0.926	1.000 (reference)
Out-of-distribution (Fashion-MNIST)	0.127	0.046

Table 8: Task-Identity under domain shift. A model trained on MNIST digit images evaluated on Fashion-MNIST clothing images shows 95.4% behavioral divergence despite identical input format.

Task-Identity of 0.046 identified a 95.4% behavioral divergence despite the identical input format. In this sample, the result suggests that Task-Identity is sensitive to semantic distribution shift, not just structural input changes.

8.3 Training Provenance

We tested whether Task-Identity can distinguish models trained on different tasks, even when evaluated on the same test set. A model trained on MNIST digit images and a model trained on Fashion-MNIST clothing images were both evaluated on the MNIST test set.

Model	Accuracy on MNIST Test	Task-Identity
MNIST-trained	0.936	1.000 (reference)
Fashion-trained	0.074	0.000

Table 9: Task-Identity between models trained on different tasks, both evaluated on the same MNIST test set. Task-Identity of 0.000 indicates 100% behavioral divergence despite identical architecture.

Task-Identity of 0.000 correctly identified complete behavioral divergence between the two models. This result supports the interpretation that Task-Identity measures learned behavior rather than architectural structure alone: two models with identical architectures trained on different tasks produce completely different behavioral fingerprints.

8.4 Progressive Noise

We tested whether Task-Identity tracks gradual degradation under increasing input noise. A model trained on clean MNIST data was evaluated across seven noise levels ranging from 0% to 30% Gaussian noise.

Noise Level	Accuracy	Task-Identity
0.00	0.936	1.000
0.05	0.933	1.000
0.10	0.922	0.999
0.15	0.880	0.993
0.20	0.793	0.948
0.25	0.695	0.864
0.30	0.614	0.780

Table 10: Task-Identity under progressively increasing Gaussian noise on MNIST. Both accuracy and Task-Identity degrade smoothly and monotonically with noise level.

Task-Identity degraded smoothly and monotonically with increasing noise, tracking accuracy degradation across the full range. In this sample, both accuracy and Task-Identity conveyed similar information, which is consistent with a degradation process that affects behavior broadly rather than targeting specific classes.

9 Discussion, Limitations, and Future Work

9.1 What This Work Shows

The central finding of this paper is that accuracy and embedding similarity can miss behavioral drift that matters for deployed classification systems, and that confusion matrix correlation provides a complementary signal that captures what those metrics leave out.

The two sharpest results illustrate this point from different angles. In the class imbalance experiment, accuracy held at 93.6% to 93.7% while Task-Identity detected a 42.4% behavioral shift. In the label space divergence experiment, embedding similarity reported 0.583 during a complete model collapse to 0% accuracy on the original task, while Task-Identity correctly read 0.000. In both cases, the failure mode was real and consequential, and the standard monitoring signal missed it.

The per-class extension sharpens the point further. In the targeted poisoning experiment, an overall Task-Identity of 0.873 concealed severe class-level compromise at 0.171 and 0.177 in exactly the two poisoned classes. In the financial lending extension, an overall score of 0.921 masked a 99.4% degradation in minority class detection across 2.26 million loan records. In both cases, the aggregate score was not just incomplete. In these cases it concealed the class-level failure that mattered most.

Not every test produced divergence between Task-Identity and traditional metrics. In the progressive noise experiment, accuracy and Task-Identity moved together, which is consistent with a setting where degradation was broad and gradual rather than selectively behavioral.

The cross-domain validation suggests that the same confusion matrix method can produce meaningful signals across computer vision, NLP, medical AI, and audio classification without domain-specific modification in the tested settings. Whether this holds for more subtle behavioral shifts across a broader range of settings is a question these results do not resolve.

9.2 Limitations

Requires labeled data. Task-Identity requires true labels for both the baseline and comparison evaluation conditions. It cannot be computed from model outputs alone. In settings where labels are expensive or unavailable at inference time, the metric cannot be applied directly.

Requires a common evaluated label set. The metric assumes that the baseline and comparison conditions can be expressed on a common label set for confusion-matrix comparison. Comparisons across genuinely incompatible label spaces are not supported by the current formulation.

Requires sufficient sample size. Confusion matrices computed from small samples produce unstable correlation estimates. The metric is most reliable when each class has enough samples to produce a stable confusion pattern. We did not systematically characterize the minimum sample size required, and this remains an open question.

Most interventions in this paper were severe. Most experiments in this paper involved a large, deliberate behavioral change: catastrophic forgetting, targeted poisoning, 4x compression, or complete domain shift. Whether Task-Identity reliably detects subtle behavioral drift, such as gradual concept drift or minor distribution shifts, is not established by these results. The progressive noise experiment is the closest to a gradual degradation test, and in that case accuracy and Task-Identity moved together rather than diverging.

Validated on a single classifier family. All experiments in this paper used scikit-learn MLPClassifier neural networks on relatively small benchmark datasets. Whether the metric behaves consistently on large-scale models, transformer architectures, or production-scale datasets is unknown.

The per-class extension is empirically motivated but not formally derived. The per-class Task-Identity analysis is supported by the poisoning and lending results, but we did not establish formal guarantees about when per-class analysis is necessary or sufficient. The moderate-to-high aggregate score masking severe class-level failure is a demonstrated phenomenon, not a guaranteed property of the metric in all settings.

9.3 Future Work

The most direct extension is to test Task-Identity on more subtle behavioral changes across larger and more diverse datasets. The current results establish that the metric works on large shifts. Whether it is reliable at smaller effect sizes is the key open question for production deployment.

A second direction is to characterize the minimum sample size needed for stable Task-Identity estimates across different numbers of classes and class distributions. This would make the metric’s applicability to specific deployment settings more concrete.

A third direction is to test the per-class extension systematically across a range of imbalance ratios and class counts, to determine how severe the imbalance must be before aggregate scores become misleading. The financial lending result suggests this can happen even at moderate imbalance ratios, but a systematic characterization would strengthen the claim.

Finally, comparing Task-Identity against a broader range of existing drift detection methods on a common benchmark would help position the metric relative to the prior art more precisely. The current comparison to accuracy monitoring and embedding similarity is illustrative but not comprehensive.

Code Availability

The Task-Identity implementation and all validation scripts used in this paper are available at <https://github.com/Wise314/task-identity>.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/906,072 (filed October 27, 2025) and U.S. Provisional Patent Application No. 63/981,437 (filed February 12, 2026). No license to implement or commercialize the described methods is granted by this publication. All rights reserved.

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